

Started on	Monday, 29 November 2021, 1:24 PM
State	Finished
Completed on	Monday, 29 November 2021, 1:24 PM
Time taken	7 secs
Marks	0/42
Grade	0 out of 100

Question 1

Not answered

Marked out of 1

Which of the following describes 'diagnostic accuracy' of a screening test?

Select one:

- ☐ Proportion of screened sample that is correctly classified by the screening test
- ☐ Proportion of screened sample with the disease
- ☐ Proportion of diseased sample classified correctly by the screening test
- ☐ Proportion of positively tested sample with the disease
- ☐ Proportion of non-diseased sample classified correctly by the screening test

Your answer is incorrect.

The term 'diagnostic accuracy' is not often used. It is replaced by much finer measures of sensitivity and specificity. Accuracy is given by all correct hits / total number of observations.

The correct answer is: Proportion of screened sample that is correctly classified by the screening test

Question 2

Not answered

Marked out of 1

Two diagnostic tests are carried out in tandem. Those who test positive for the first test are further tested using the second test in a serial manner. If the two tests are considered as a single diagnostic battery for final results, which of the following is correct?

Select one:

- ☐ The sensitivity and specificity are unchanged
- ☐ The sensitivity and specificity depends on the order of individual test application
- ☐ The battery has higher sensitivity than the individual tests
- ☐ The battery has higher specificity than individual tests
- ☐ The battery has lower sensitivity and specificity than individual tests



Your answer is incorrect.

When two sequential tests are used, and those who test positive by the first test are brought in for the second test, there is a net loss in sensitivity, but a net gain in specificity, compared with either test alone. However, when two simultaneous tests are used, there is a net gain in sensitivity and a net loss of specificity, compared with either test alone.

The correct answer is: The battery has higher specificity than individual tests

Question 3

Not answered

Marked out of 1

Which of the following expression refers to the likelihood that a positive test is found in a person with the disease, as opposed to the one without the disease?

Select one:

- ☐ specificity/ (1-sensitivity)
- ☐ (1-Sensitivity)/ specificity
- ☐ Sensitivity / specificity
- ☐ (1-Sensitivity)/ (1-specificity)
- ☐ Sensitivity/ (1-specificity)

Your answer is incorrect.

The likelihood ratio of a positive test (LR+) is the ratio of the likelihood (probability) of a positive test result in the population with disease divided by the likelihood of a positive test result in the population without disease. A simple way of calculating this is using the expression $LR+ = \text{Sensitivity} / (1 - \text{specificity})$

The correct answer is: Sensitivity/ (1-specificity)

Question 4

Not answered

Marked out of 1

A dementia screening test is used in two similar age-matched populations (primary care attenders vs. hospital outpatient attenders) under identical test conditions. But the probability of having the disease when having a positive result among hospital attenders is higher than that among primary care attenders. The most likely explanation is

Select one:

- ☐ The specificity of the test is higher among primary care attenders
- ☐ The specificity of the test is lower among hospital attenders
- ☐ It is not possible to determine the cause without repeating the screening
- ☐ The prevalence of dementia is higher among hospital attenders
- ☐ The prevalence of dementia is higher among primary care attenders

Your answer is incorrect.

Higher prevalence in the screened population can lead to a marked increase in the positive predictive value using the same test - this gain plateaus after reaching a certain degree of higher prevalence.

The correct answer is: The prevalence of dementia is higher among hospital attenders

Question 5

Not answered

Marked out of 1

Two researchers debate the usefulness of a new laboratory tests that claim to detect Wernicke's encephalopathy. Dr. A uses the usual diagnostic examination (physical examination + MRI) which has a sensitivity of 90% and a specificity of 96%. Dr. B uses the new test, which is 96% sensitive and 96% specific. If 200 patients undergo investigation with both the new test and established diagnostic method, which of the following is correct?

Select one:

- ☐ The prevalence of Wernicke's encephalopathy is needed to determine which researcher will correctly identify the larger number of people with the disease
- ☐ Dr. A will correctly identify more people with Wernicke's encephalopathy than Dr. B
- ☐ Dr. A will correctly identify fewer people with Wernicke's encephalopathy than Dr. B
- ☐ Dr A will not wrongly diagnose any subject
- ☐ Dr. A will correctly identify more people without Wernicke's encephalopathy than Dr. B

Your answer is incorrect.

Both tests have equal specificity. Only sensitivity varies. In this case, the test with higher sensitivity will correctly identify more people with the disease than the test with lower sensitivity.

The correct answer is: Dr. A will correctly identify fewer people with Wernicke's encephalopathy than Dr. B

Question 6

Not answered

Marked out of 1

A two-item screening interview and a laboratory test were administered to 500 persons with suspected alcohol problems, of whom 300 were actually found to have a diagnosis of alcohol dependence. (see the attached image). Compared with the two-item interview, the lab test has

Two-item Interview		
	DSM Alcohol dependence	
Result	Present	Absent
Positive	240	40
Negative	60	160

Lab test		
	DSM Alcohol dependence	
Result	Present	Absent
Positive	270	60
Negative	30	140

Select one:

- ☐ Less sensitivity and less specificity
- ☐ More sensitivity and less specificity
- ☐ More sensitivity and more specificity
- ☐ Same sensitivity and specificity
- ☐ Less sensitivity and more specificity

Your answer is incorrect.

Two item test: sensitivity = $240/300 = 80\%$. Specificity = $160/200 = 80\%$. Lab test: sensitivity = $270/300 = 90\%$. Specificity = $140/200 = 70\%$. Hence the lab test has more sensitivity but less specificity.

The correct answer is: More sensitivity and less specificity

Question **7**

Not answered

Marked out of 1

A pilot project carried out by DVLA uses a weekly cut-off of 21 units to define alcohol misuse. If this cut-off is reduced to 14 units which of the following might happen?

Select one:

- ☐ Increase in false positives
- ☐ Increase in positive predictive value
- ☐ Decrease in true positives
- ☐ Decrease in sensitivity
- ☐ Decrease in false positives

Your answer is incorrect.

The dilemma involved in deciding whether to set a high cutoff or a low cutoff rests in the problem of the false positives and the false negatives that result from the testing. If a high cutoff is set, then the sensitivity will be low while the specificity will be high. If a low cut-off is chosen, a large proportion false positives result. This will increase the sensitivity, but the specificity will drop down.

The correct answer is: Increase in false positives

Question 8

Not answered

Marked out of 1

A retired school teacher is having problems with his memory. He is seen by a trainee doctor at home who administers MMSE and reports a score of 27. Which of the following test characters is most meaningful in this situation to reassure absence of significant cognitive impairment amounting to dementia

Select one:

- ☐ Negative predictive value
- ☐ Sensitivity
- ☐ Specificity
- ☐ Accuracy
- ☐ Positive predictive value

Your answer is incorrect.

Every one of us, after getting results of a test for a patient, want to know only one thing - "Given this test result, what is the likelihood that the patient has the disease?" Sensitivity and specificity are characteristics of the test - not of the patient with a particular test result. The most useful measure to answer the above is the predictive value. This is affected by two factors: the prevalence of the disease in the population tested and, if the disease is rare, the specificity of the test being used. For a negative test result, the Negative predictive value will be more useful.

The correct answer is: Negative predictive value

Question 9

Not answered

Marked out of 1

A 75-year-old man presents with cognitive difficulties and has a score of 24 in MMSE. It is reported that MMSE scores ≥ 25 has a sensitivity of 80% and specificity of 90% to detect dementia. It is also known that the prevalence of dementia at the age of 75 is around 10%. What is the probability that this man actually has dementia?

Select one:

- ☐ 0.47
- ☐ 0.8
- ☐ 0.9
- ☐ 0.22
- ☐ 0.6

Your answer is incorrect.

To determine the posttest probability, we must first determine the post-test odds. Post-test odds are given by $LR \times \text{pretest odds}$. Pretest odds can be calculated from the prevalence which is, in fact, the pretest probability. Here likelihood ratio for positive test is given by $LR+ = \text{sensitivity} / (1 - \text{specificity}) = 8$. Posttest odds = $1/9 \times 8 = 8/9$. Post-test probability = $8/17 = 0.47$

The correct answer is: 0.47

Question 10

Not answered

Marked out of 1

Which of the following properties of a screening test is not affected by the prevalence rate of the disease?

Select one:

- ☐ Negative predictive value
- ☐ Likelihood ratio
- ☐ Post test probability
- ☐ Positive predictive value
- ☐ Pre test probability

Your answer is incorrect.

The LR+ is simply the quotient of the sensitivity of the false positive rate. Both of these terms are not influenced by the prevalence; therefore, neither is the LR+.

The correct answer is: Likelihood ratio

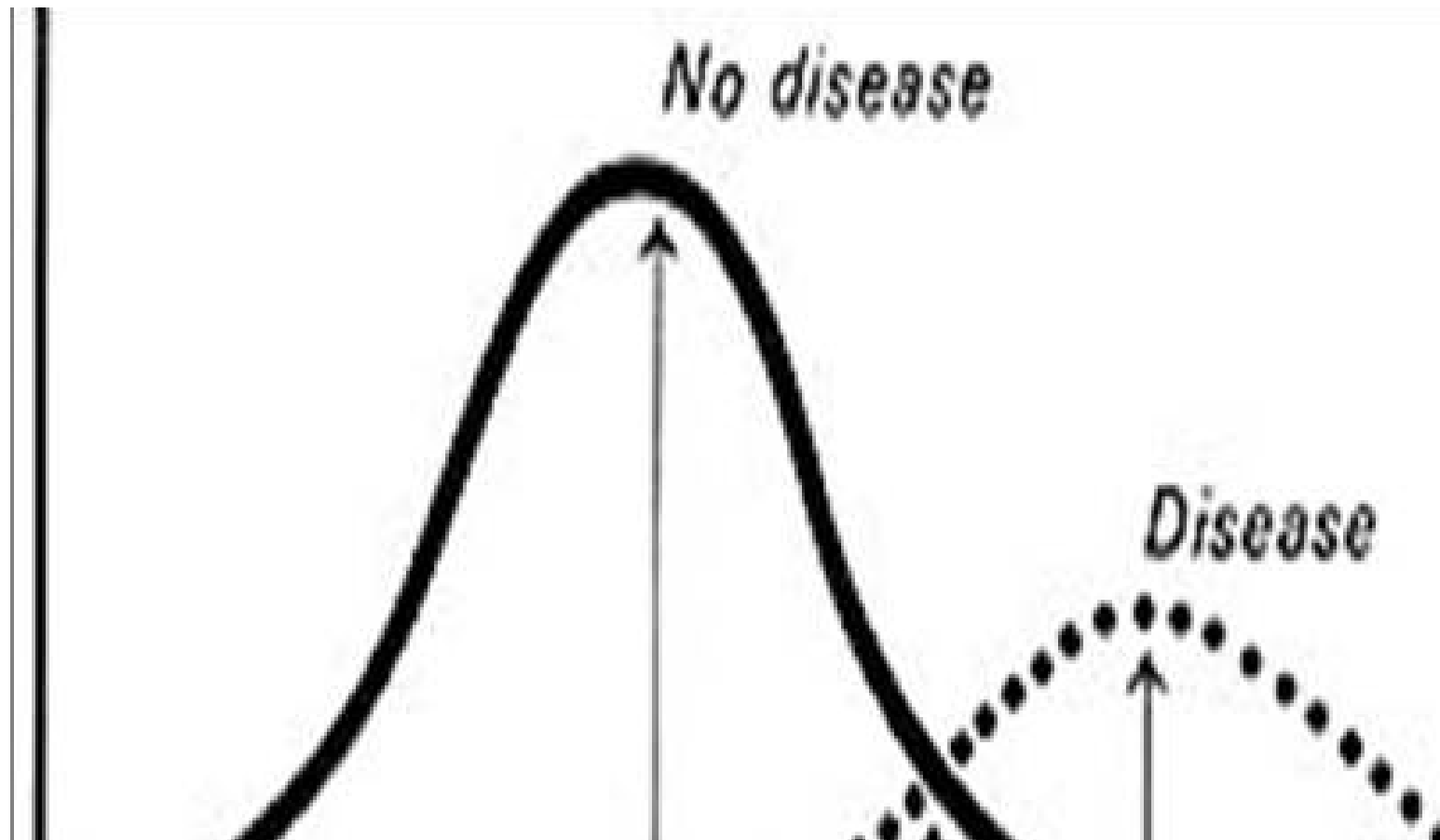


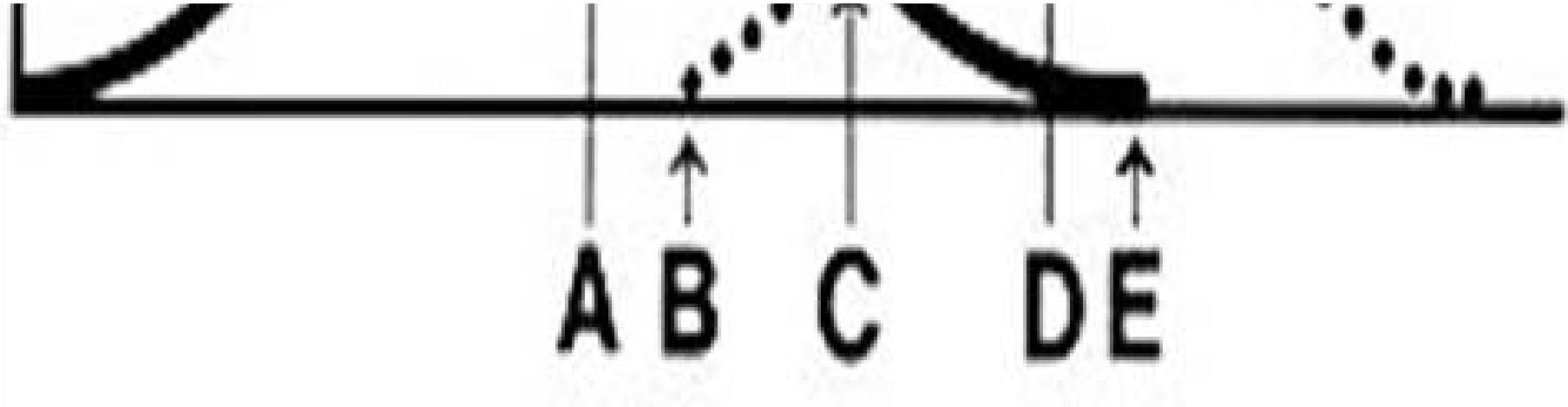
Question 11

Not answered

Marked out of 1

A cheap screening test has been devised for vCJD through analysis of the degree of certain polymorphism in hair follicles or skin scrapings. The population distribution of this polymorphism is shown in the attached graph. Assuming suitable curative treatment is available for vCJD if discovered early, which is the most appropriate cutoff point to be chosen to define a positive test result?





Select one:

- ☐ B
- ☐ C
- ☐ A
- ☐ E
- ☐ D

Your answer is incorrect.

The dilemma involved in deciding whether to set a high cutoff or a low cutoff rests in the problem of the false positives and the false negatives that result from the testing. For a deadly disease that has a curative treatment if discovered early, we must avoid false negatives even if it comes at a cost of higher false positives.

The correct answer is: B

Question 12

Not answered

Marked out of 1

You are worried that a bipolar patient treated with valproate for acute mania in your ward may be pregnant. She has a chaotic lifestyle and has no account of last menstrual period. The nurses have carried out a dip-stick test for pregnancy and reported a negative result, but you are not convinced. You manage to persuade your patient to undergo another pregnancy test. Assuming that urine dip-test for pregnancy has a sensitivity of 90% and a specificity of 88% what is the probability that your patient, if pregnant, will have two consecutive negative pregnancy tests?

Select one:

- ☐ 90/100
- ☐ 88/100
- ☐ 1/100
- ☐ 44/100
- ☐ 180/100

Your answer is incorrect.

Assuming the two test results to be independent of each other, we can multiply the probability of a negative test result in 'truly diseased' to achieve the probability at the end of consecutive negative tests. The probability of a negative test if there is pregnancy is given by false negative rate = (Test negative)/ (disease positive). This can be obtained from 1-sensitivity. Here it is $(1 - 0.9 = 0.1)$. The probability of two negative consecutive tests is $(0.1)(0.1) = 0.01 = 1/100$.

The correct answer is: 1/100

Question 13

Not answered

Marked out of 1

To evaluate the performance of a new diagnostic questionnaire the questionnaire developer checks it out on 100 known cases of the disease for which the scale was designed, and on 200 controls known to be free of the disease. Ninety of the cases yield positive tests, as do 30 of the controls. Based on these data, the specificity and false positive error rates of the new questionnaire are, respectively

Select one:

- ☐ 75% and 10%
- ☐ 85% and 25%
- ☐ 85% and 10%
- ☐ 90% and 15%
- ☐ 85% and 15%



Your answer is incorrect.

Draw a 2X2 table as shown below.

diagtest/goldstd	disease +	disease -	total
	cases	controls	

From this, specificity = $170/200 = 85\%$ and false positive rate is always $(1 - \text{specificity}) = 15\%$

The correct answer is: 85% and 15%

Question 14

Not answered

Marked out of 1

Based on a local audit and survey result you estimate that a patient in your catchment area has one in four chance of having panic disorder. You want to use a recently published Panic Disorder Instrument (PDI) with a sensitivity of 95% and a specificity of 90% to diagnose a patient. The result comes back positive. Based on all the above information the chance your patient really has the disease is nearly

Select one:

- ☐ 0.95
- ☐ 0.9
- ☐ 1
- ☐ 0.7
- ☐ 0.6

Your answer is incorrect.

Post-test odds are given by $LR \times \text{pretest odds}$. Pretest odds can be calculated from the prevalence that is, in fact, the pretest probability. For the probability of $1/4$, the odds are $1/3$. Here likelihood ratio for positive test is given by $LR+ = \text{sensitivity} / (1 - \text{specificity}) = 9.5$. Posttest odds = $1/3 \times 9.5 = 3.16$. Post-test probability = $3.16 / 4.16 = 0.75$

The correct answer is: 0.7

Question 15

Not answered

Marked out of 1

An evaluation of a new screening questionnaire for depression in primary care is conducted. The most important single feature of this questionnaire that would encourage you to use it is:

Select one:

- ☐ B. It has sensitivity of 80%
- ☐ C. It has a specificity of 60%
- ☐ A. It has Positive Predictive Value of 20%
- ☐ E. It is validated in many languages
- ☐ D. It is brief



Your answer is incorrect.

A PPV of 20% is too low for a screening tool. Having a sensitivity of 80% suggests that most depressed patients will be screened in by the questionnaire, thus reducing number of missed cases, which is probably the main purpose of screening tools in primary care.

The correct answer is: B. It has sensitivity of 80%

Question 16

Not answered

Marked out of 1

Using a model derived from various scoring methods of MRI brain images, Ahearn et al. reported the following data how MRI could be used to predict suicide attempts. 20 patients without a history of suicide attempt and 20 patients with a history of suicide attempt were scanned. MRI model was able to predict the absence of suicide history correctly in 19 of those who had never attempted suicide. The model also predicted 16 to have suicide history out of 20 who had such a history. The percentage of those with a positive MRI result who actually have attempted suicide is

Select one:

- ☐ 4/20 %
- ☐ 19/20 %
- ☐ 16/20 %
- ☐ 1/17 %
- ☐ 16/17 %

Your answer is incorrect.

It is very important to derive the following to answer the above question. Please try to make up a 2X2 table using the following instruction.

Firstly - the gold standard test is having an actual history of suicide attempt. The diagnostic test evaluated is the MRI prediction.

Secondly - there are 20 "attempters" and 20 "non-attempters" at inception. Of these 19 true negatives and 16 true positives. Hence total MRI positive is $16 + 1 = 17$; total MRI negative is $19 + 4 = 23$. Out of 17 who are MRI positive, 16 are actually suicide attempt positive. So the answer is 16/17 %. This is the positive predictive value.

TEST/GOLD STD	▼	attempter	▼	non-attempter	▼	total	▼
MRI + (predicted to have suicide attempt)		16		1		17	
MRI - (predicted to NOT have suicide attempt)		4		19		23	
TOTAL		20		20		40	

The correct answer is: 16/17 %



Question 17

Not answered

Marked out of 1

Using a model derived from various scoring methods of MRI brain images, Ahearn et al. reported the following data how MRI could be used to predict suicide attempts. 20 patients without a history of suicide attempt and 20 patients with a history of suicide attempt were scanned. MRI model was able to predict the absence of suicide history correctly in 19 of those who had never attempted suicide. The model also predicted 16 to have suicide history out of 20 who actually had such a history. The accuracy of the MRI test in predicting suicide is given by

	Predicted to have absence of suicide attempts (Number of patients)	Predicted to have a history of suicide attempts (Number of patients)
Absence of suicide attempt	19 (95%)	1(5%)
Presence of suicide attempt	4(20%)	16(80%)

Select one:

☐ 16/17

☐ 19/20

- ☐ 16/40
- ☐ (1/17)
- ☐ 35/40

Your answer is incorrect.

It is very important to derive the following to answer the above question. Please try to make up a 2X2 table using the following instruction. Here the gold standard is having an actual history of suicide attempt. The diagnostic test evaluated is the MRI prediction. There are 20 total positives and 20 total negatives at inception. Of these 19 true negatives and 16 true positives. Hence total MRI positive is $16 + 1 = 17$; total MRI negative is $19 + 4 = 23$. Accuracy is given by total correct hits/ total sample studied = $(19 + 16) / (20+20) = 35/40$

The correct answer is: 35/40



Question **18**

Not answered

Marked out of 1

Using a model derived from various scoring methods of MRI brain images, Ahearn et al. reported the following data how MRI could be used to predict suicide attempts. 20 patients without a history of suicide attempt and 20 patients with a history of suicide attempt were scanned. MRI model was able to predict the absence of suicide history correctly in 19 of those who had never attempted suicide. The model also predicted 16 to have suicide history out of 20 who actually had such a history. The probability of the MRI test to predict suicide attempts among those who really attempted suicide is given by

Select one:

- ☐ 16/17
- ☐ (4/20)
- ☐ 19/23
- ☐ 16/20
- ☐ (1/17)

Your answer is incorrect.

TEST/GOLD STD	▼ attempter ▼	non-attempter ▼	total ▼
MRI + (predicted to have suicide attempt)	16	1	17
MRI - (predicted to NOT have suicide attempt)	4	19	23
TOTAL	20	20	40

It is very important to derive the following to answer the above question. Please try to make up a 2X2 table using the following instruction.

Firstly - The gold standard test is the actual history of suicide attempt. The diagnostic test evaluated is the MRI prediction. So the table in the question has rows and columns interchanged.

Secondly - There are 20 total positives and 20 total negatives at inception. Of these 19 true negatives and 16 true positives.

The probability of the MRI test to predict suicide attempts among those who really attempted suicide = sensitivity = $16/20$.

The correct answer is: $16/20$



Question 19

Not answered

Marked out of 1

Using a model derived from various scoring methods of MRI brain images, Ahearn et al. reported the following data how MRI could be used to predict suicide attempts. 20 patients without a history of suicide attempt and 20 patients with a history of suicide attempt were scanned. MRI model was able to predict the absence of suicide history correctly in 19 of those who had never attempted suicide. The model also predicted 16 to have suicide history out of 20 who actually had such a history. The probability of the MRI test to correctly predict absence of suicide attempts among those who never attempted suicide is given by

	Predicted to have absence of suicide attempts (Number of patients)	Predicted to have a history of suicide attempts (Number of patients)
Absence of suicide attempt	19 (95%)	1(5%)
Presence of suicide attempt	4(20%)	16(80%)

Select one:

☐ 1/17.

- ☐ 19/20
- ☐ (1/20)
- ☐ 16/20
- ☐ 16/17

Your answer is incorrect.

It is very important to derive the following to answer the above question. Please try to make up a 2X2 table using the following instruction. Here the gold standard is having an actual history of suicide attempt. The diagnostic test evaluated is the MRI prediction. There are 20 total positives and 20 total negatives at inception. Of these 19 true negatives and 16 true positive. The probability of the MRI test to predict the absence of suicide attempts among those who never had attempted suicide = specificity = 19/20.

The correct answer is: 19/20



Question 20

Not answered

Marked out of 1

Using a model derived from various scoring methods of MRI brain images, Ahearn et al. reported the following data how MRI could be used to predict suicide attempts. 20 patients without a history of suicide attempt and 20 patients with a history of suicide attempt were scanned. MRI model was able to predict the absence of suicide history correctly in 19 of those who had never attempted suicide. The model also predicted 16 to have suicide history out of 20 who actually had such a history. How many times more likely is that one will find a positive MRI test in those who really attempted suicide versus those who did not attempt it?

	Predicted to have absence of suicide attempts (Number of patients)	Predicted to have a history of suicide attempts (Number of patients)
Absence of suicide attempt	19 (95%)	1(5%)
Presence of suicide attempt	4(20%)	16(80%)

Select one:

☐ 16

- ☐ 0.16
- ☐ 8
- ☐ 0.8
- ☐ 5

Your answer is incorrect.

Here we are asked to find likelihood ratio of the positive test. It is very important to derive the following to answer the above question. Please try to make up a 2X2 table using the following instruction. Here the gold standard is having an actual history of suicide attempt. The diagnostic test evaluated is the MRI prediction. There are 20 total positives and 20 total negatives at inception. Of these 19 true negatives and 16 true positive. $LR+ = \text{likelihood of testing positive when diseased} / \text{likelihood of testing positive when not diseased} = (16/20) / (1/20) = 16$. Another way of calculating this is $LR+ = \text{sensitivity} / (1 - \text{specificity}) = (16/20) / (1 - [19/20]) = 16$.

The correct answer is: 16



Question 21

Not answered

Marked out of 1

Which of the following corresponds to a prevalence of a disease in the studied population for which a diagnostic test is evaluated?

Select one:

- ☐ Post test probability
- ☐ Pre test probability
- ☐ Pre test odds
- ☐ Likelihood ratio of positive test
- ☐ Positive predictive value

Your answer is incorrect.

Pre test probability is same as the prevalence in the studied population. One must adjust the likelihood ratio according to this value while applying a new scale or test.

The correct answer is: Pre test probability



Question 22

Not answered

Marked out of 1

The sensitivity, specificity, positive, and negative predictive values of GHQ-12 in detecting post-miscarriage psychiatric morbidity is given by the below table. Which of the following is true if the cut off is decided at GHQ>2?

Select one:

- ☐ None of the above
- ☐ Less people without significant morbidity will be diagnosed wrongly
- ☐ Everyone without the disease will be excluded correctly
- ☐ If the test is positive then the patient has 100% chance of being ill
- ☐ Everyone with the disease will be diagnosed correctly

Your answer is incorrect.

At GHQ > 2, Sensitivity = 100%; Negative predictive value = 100%. But specificity is 77% and PPV = 14%. Sensitivity 100% means true positives = total diseased. So there are no false negatives. The same can be inferred from NPV = true negative / total test negative = 100% i.e. there are no false negatives. Consider statement A. 'Everyone without the disease will be excluded correctly' - this means there will be no false positives. This is not the inference from the above deduction. Consider statement B. 'If the test is positive then the patient has 100% chance of being ill' - This means positive predictive value = 100%. This is not true either. Consider statement C. 'Everyone with the disease will be diagnosed correctly' - this means that 'total diseased = true positives'. This can be deducted from the above values of sensitivity. Consider statement D. 'Less people without significant morbidity will be diagnosed wrongly'. This means 'false positive' level is very low or near zero. This is not the case with above values. Hence, the answer is C

The correct answer is: Everyone with the disease will be diagnosed correctly

Question **23**

Not answered

Marked out of 1

The sensitivity, specificity, positive, and negative predictive values of GHQ-12 in detecting post-miscarriage psychiatric morbidity is given by the below table. At which cut off point does a patient who turns positive have the highest probability of being truly diseased?

Select one:

- ☐ >6
- ☐ >4
- ☐ >7
- ☐ >5
- ☐ >2



Your answer is incorrect.

This question seeks the point at which the positive predictive value moves closest to 100%. This happens when the cut-off is 7.

The correct answer is: >7

Question 24

Not answered

Marked out of 1

A researcher compares two diagnostic tests for alcohol dependence, one of which is considered to be a 'gold standard'. What is the single most important issue to consider when assessing the validity of such a study?

Select one:

- ☐ The sensitivity and specificity of the test.
- ☐ The test-retest reliability of the 'gold standard'.
- ☐ The impact factor of the journal
- ☐ Whether the test and 'gold standard' were applied independently
- ☐ The clinical diagnostic ability of the researcher



Your answer is incorrect.

It is of utmost importance that both gold standard test and screening tests are applied independent of each other in a cross-sectional study that evaluates diagnostic tests.

The correct answer is: Whether the test and 'gold standard' were applied independently

Question 25

Not answered

Marked out of 1

The Warwick-Edinburgh Mental Well-Being Scale (WEMWBS) was developed to meet the demand for instruments to measure mental well-being. It comprises 14 positively phrased Likert-style items and fulfils classic criteria for scale development. The authors report the internal construct validity of WEMWBS from the perspective of the Rasch measurement model. The model was applied to data collected from 779 respondents (aged 16 –74 (average 41.9) yrs) in Wave 12 (Autumn 2006) of the Scottish Health Education Population Survey. Initial fit to model expectations was poor. The items 'I've been feeling good about myself', 'I've been interested in new things' and 'I've been feeling cheerful' all showed significant misfit to model expectations and were deleted. This led to a marginal improvement in fit to the model. After further analysis, more items were deleted, and a strict unidimensional seven-item scale (the Short Warwick Edinburgh Mental Well-Being Scale (SWEMWBS)) was resolved. Many items deleted because of misfit with model expectations showed considerable bias for gender. Two retained items also demonstrated a bias for gender but, at the scale level, cancelled out. One further retained item 'I've been feeling optimistic about the future' showed bias for age. The correlation between the 14 item and 7 item versions was 0.954. The authors concluded that the short 7-item version of WEMWBS was found to satisfy the strict unidimensionality expectations of the Rasch model, and be largely free of bias. This scale, SWEMWBS, provides a raw score-interval scale transformation for use in parametric procedures. In terms of face validity, SWEMWBS presents a more restricted view of mental well-being than the 14 item WEMWBS, with most items representing aspects of psychological and eudemonic well-being, and few covering hedonic well-being or affect. However, robust measurement properties combined with brevity make SWEMWBS preferable to WEMWBS at present for monitoring mental well-being in populations. Where face validity is an issue, the authors favour continuing to collect data on the full 14 item WEMWBS. Stewart-Brown et al., Health and Quality of Life Outcomes 2009, 7:15. This study describes the development of a questionnaire. What is the application for which this questionnaire is being developed?

Select one:

- ☐ To screen for mental disorders among respondents in general population
- ☐ To assess mental wellbeing of respondents in general population
- ☐ To exclude mental disorders among respondents in general population
- ☐ To measure the wellbeing of psychiatric inpatients
- ☐ To measure mental wellbeing of children receiving health education

Your answer is incorrect.

This questionnaire purports to measure the mental wellbeing of general population respondents

The correct answer is: To assess mental wellbeing of respondents in general population



Question 26

Not answered

Marked out of 1

The Warwick-Edinburgh Mental Well-Being Scale (WEMWBS) was developed to meet the demand for instruments to measure mental well-being. It comprises 14 positively phrased Likert-style items and fulfils classic criteria for scale development. The authors report the internal construct validity of WEMWBS from the perspective of the Rasch measurement model. The model was applied to data collected from 779 respondents (aged 16 –74 (average 41.9) yrs) in Wave 12 (Autumn 2006) of the Scottish Health Education Population Survey. Initial fit to model expectations was poor. The items 'I've been feeling good about myself', 'I've been interested in new things' and 'I've been feeling cheerful' all showed significant misfit to model expectations and were deleted. This led to a marginal improvement in fit to the model. After further analysis, more items were deleted, and a strict unidimensional seven-item scale (the Short Warwick Edinburgh Mental Well-Being Scale (SWEMWBS)) was resolved. Many items deleted because of misfit with model expectations showed considerable bias for gender. Two retained items also demonstrated a bias for gender but, at the scale level, cancelled out. One further retained item 'I've been feeling optimistic about the future' showed bias for age. The correlation between the 14 item and 7 item versions was 0.954. The authors concluded that the short 7-item version of WEMWBS was found to satisfy the strict unidimensionality expectations of the Rasch model, and be largely free of bias. This scale, SWEMWBS, provides a raw score-interval scale transformation for use in parametric procedures. In terms of face validity, SWEMWBS presents a more restricted view of mental well-being than the 14 item WEMWBS, with most items representing aspects of psychological and eudemonic well-being, and few covering hedonic well-being or affect. However, robust measurement properties combined with brevity make SWEMWBS preferable to WEMWBS at present for monitoring mental well-being in populations. Where face validity is an issue, the authors favour continuing to collect data on the full 14 item WEMWBS. Stewart-Brown et al., Health and Quality of Life Outcomes 2009, 7:15. While developing this scale, the authors deleted three items - 'I've been feeling good about myself', 'I've been interested in new things' and 'I've been feeling cheerful'. Deleting these three items has improved which one of the following?

Select one:

- ☐ Construct validity
- ☐ Inter-rater reliability
- ☐ Predictive validity
- ☐ Correlation with age
- ☐ Stability of the scores across time

Your answer is incorrect.

The 7-item version is likely to have higher construct validity as the items that had poor fit with the model of mental wellbeing have been removed. This deletion per se will not have any discernible effect on the reliability metrics. The precis has no data to support or refute a possible effect on other measures (predictive validity or stability) listed in the question.

The correct answer is: Construct validity



Question 27

Not answered

Marked out of 1

The Warwick-Edinburgh Mental Well-Being Scale (WEMWBS) was developed to meet the demand for instruments to measure mental well-being. It comprises 14 positively phrased Likert-style items and fulfils classic criteria for scale development. The authors report the internal construct validity of WEMWBS from the perspective of the Rasch measurement model. The model was applied to data collected from 779 respondents (aged 16 –74 (average 41.9) yrs) in Wave 12 (Autumn 2006) of the Scottish Health Education Population Survey. Initial fit to model expectations was poor. The items 'I've been feeling good about myself', 'I've been interested in new things' and 'I've been feeling cheerful' all showed significant misfit to model expectations and were deleted. This led to a marginal improvement in fit to the model. After further analysis, more items were deleted, and a strict unidimensional seven-item scale (the Short Warwick Edinburgh Mental Well-Being Scale (SWEMWBS)) was resolved. Many items deleted because of misfit with model expectations showed considerable bias for gender. Two retained items also demonstrated a bias for gender but, at the scale level, cancelled out. One further retained item 'I've been feeling optimistic about the future' showed bias for age. The correlation between the 14 item and 7 item versions was 0.954. The authors concluded that the short 7-item version of WEMWBS was found to satisfy the strict unidimensionality expectations of the Rasch model, and be largely free of bias. This scale, SWEMWBS, provides a raw score-interval scale transformation for use in parametric procedures. In terms of face validity, SWEMWBS presents a more restricted view of mental well-being than the 14 item WEMWBS, with most items representing aspects of psychological and eudemonic well-being, and few covering hedonic well-being or affect. However, robust measurement properties combined with brevity make SWEMWBS preferable to WEMWBS at present for monitoring mental well-being in populations. Where face validity is an issue, the authors favour continuing to collect data on the full 14 item WEMWBS. Stewart-Brown et al., Health and Quality of Life Outcomes 2009, 7:15. Which of the following measures, if obtained, could demonstrate that the scores of the short version of WEMWBS could remain stable over a period of 2 weeks?

Select one:

- ☐ Split half reliability
- ☐ Predictive validity
- ☐ Test-retest reliability
- ☐ Construct validity
- ☐ Internal consistency

Your answer is incorrect.

The test-retest reliability is a method of testing the stability and reliability of an instrument over time.

The correct answer is: Test-retest reliability



Question 28

Not answered

Marked out of 1

The Warwick-Edinburgh Mental Well-Being Scale (WEMWBS) was developed to meet the demand for instruments to measure mental well-being. It comprises 14 positively phrased Likert-style items and fulfils classic criteria for scale development. The authors report the internal construct validity of WEMWBS from the perspective of the Rasch measurement model. The model was applied to data collected from 779 respondents (aged 16 –74 (average 41.9) yrs) in Wave 12 (Autumn 2006) of the Scottish Health Education Population Survey. Initial fit to model expectations was poor. The items 'I've been feeling good about myself', 'I've been interested in new things' and 'I've been feeling cheerful' all showed significant misfit to model expectations and were deleted. This led to a marginal improvement in fit to the model. After further analysis, more items were deleted, and a strict unidimensional seven-item scale (the Short Warwick Edinburgh Mental Well-Being Scale (SWEMWBS)) was resolved. Many items deleted because of misfit with model expectations showed considerable bias for gender. Two retained items also demonstrated a bias for gender but, at the scale level, cancelled out. One further retained item 'I've been feeling optimistic about the future' showed bias for age. The correlation between the 14 item and 7 item versions was 0.954. The authors concluded that the short 7-item version of WEMWBS was found to satisfy the strict unidimensionality expectations of the Rasch model, and be largely free of bias. This scale, SWEMWBS, provides a raw score-interval scale transformation for use in parametric procedures. In terms of face validity, SWEMWBS presents a more restricted view of mental well-being than the 14 item WEMWBS, with most items representing aspects of psychological and eudemonic well-being, and few covering hedonic well-being or affect. However, robust measurement properties combined with brevity make SWEMWBS preferable to WEMWBS at present for monitoring mental well-being in populations. Where face validity is an issue, the authors favour continuing to collect data on the full 14 item WEMWBS. Stewart-Brown et al., Health and Quality of Life Outcomes 2009, 7:15. When comparing the short 7 item version of WEMWBS with the full 14 item version, which of the following is a correct statement?

Select one:

- ☐ 7-item version has poor correlation with 14-item version
- ☐ 14-item version cannot be used to acquire data for parametric statistical tests
- ☐ 14-item version is free of gender bias
- ☐ 7-item version is easier to use
- ☐ 14-item version has less face validity

Your answer is incorrect.

The 7-item version is likely to have higher construct validity as the items that had poor fit with the model of mental wellbeing have been removed. Some such removed items, on causal inspection, appear to describe what most people would agree as an aspect of mental wellbeing. So the short version has lost some face validity. Nevertheless, it is easier to use due to its brevity. Furthermore, as the items showing gender/age bias have been removed, the short version is less likely to be confounded by the gender or age of the respondent.

The correct answer is: 7-item version is easier to use



Question 29

Not answered

Marked out of 1

The Warwick-Edinburgh Mental Well-Being Scale (WEMWBS) was developed to meet demand for instruments to measure mental well-being. It comprises 14 positively phrased Likert-style items and fulfils classic criteria for scale development. The authors report the internal construct validity of WEMWBS from the perspective of the Rasch measurement model. The model was applied to data collected from 779 respondents (aged 16 –74 (average 41.9) yrs) in Wave 12 (Autumn 2006) of the Scottish Health Education Population Survey. Initial fit to model expectations was poor. The items 'I've been feeling good about myself', 'I've been interested in new things' and 'I've been feeling cheerful' all showed significant misfit to model expectations and were deleted. This led to a marginal improvement in fit to the model. After further analysis, more items were deleted, and a strict unidimensional seven-item scale (the Short Warwick Edinburgh Mental Well-Being Scale (SWEMWBS)) was resolved. Many items deleted because of misfit with model expectations showed considerable bias for gender. Two retained items also demonstrated a bias for gender but, at the scale level, cancelled out. One further retained item 'I've been feeling optimistic about the future' showed bias for age. The correlation between the 14 item and 7 item versions was 0.954. The authors concluded that the short 7-item version of WEMWBS was found to satisfy the strict unidimensionality expectations of the Rasch model, and be largely free of bias. This scale, SWEMWBS, provides a raw score-interval scale transformation for use in parametric procedures. In terms of face validity, SWEMWBS presents a more restricted view of mental well-being than the 14 item WEMWBS, with most items representing aspects of psychological and eudemonic well-being, and few covering hedonic well-being or affect. However, robust measurement properties combined with brevity make SWEMWBS preferable to WEMWBS at present for monitoring mental well-being in populations. Where face validity is an issue, the authors favour continuing to collect data on the full 14 item WEMWBS. Stewart-Brown et al., Health and Quality of Life Outcomes 2009, 7:15. What can be said about the scores derived from the short version WEMWBS?

Select one:

- ☐ Mann-Whitney U Test cannot be used to compare the total scores from 2 groups
- ☐ T tests cannot be used to compare the total scores from 2 groups
- ☐ When considering the total scores within a group, the mean cannot be same as the median
- ☐ The total scores do not have meaningful units
- ☐ The values of total scores can vary anywhere between 0 and infinity

Your answer is incorrect.

The authors have indicated that the scores from the short version scale are interval measures. As such, interval measures can be compared by using either T-tests (parametric) or Mann Whitney U Tests depending on whether the assumptions of normality hold true. Unlike continuous ratio scores, an interval score does not have meaningful units. Being a summed score, it can only vary within a defined range though it is possible to obtain mean and standard deviation measures.

The correct answer is: The total scores do not have meaningful units



Question 30

Not answered

Marked out of 1

Using the table below, which reports results of a UK-based study assessing the CAGE to identify people with alcohol dependence, answer questions that follow. If the criterion for defining alcohol misuse was answering yes to at least 3 questions, what is the sensitivity of the CAGE?

Select one:

- ☐ 20%
- ☐ 36%
- ☐ 60%
- ☐ 16%
- ☐ 74%

Your answer is incorrect.

From the table provided in the question, create a new 'standard table' that can be used to calculate the properties of a diagnostic instrument as below. Note that the term Test-Positive for a cut-off of 3 means that the individual is scoring either 3 or 4 in the CAGE. There are 9 such individuals (see the 'Total' column in the original table in the question). This number 9 must be entered in (A+B) of the new table. The remaining 66 will be in (C+D). When you divide the test positives into Reference Standard 'positive' and 'negative', all 9 of the test positives appear to be in the 'reference standard positive' category in the original table. Place them in cell A (or Disease positive in the new table). Similarly you will find the value 16 will enter cell C, 0 will enter cell B, and 50 will enter cell D. Please complete the table as shown here. Now sensitivity = $A/A+C = 9/25 \times 100 = 36\%$. Note that in similar fashion you could calculate sensitivity and specificity for each of the various cut-off values of the CAGE. Please keep in mind that the values of the 'total' column in the original table should not be used as such, but must be used in a cumulative fashion according to the cut-off in question. (A cut-off categorises people who score equal to and more than that score in the same section = Test positive group)

what score in the same section? (100 positive group)

The correct answer is: 36%



Question 31

Not answered

Marked out of 1

Using the table below, which reports results of a UK-based study assessing the CAGE to identify people with alcohol dependence, answer questions that follow. Using a score of 4 instead of 3 or more to indicate a positive test result will produce which of the following effect?

Select one:

- ☐ An increase in both sensitivity and specificity
- ☐ A decrease in sensitivity but increase in positive predictive value.
- ☐ A decrease in sensitivity but increase in specificity.
- ☐ A decrease in both sensitivity and positive predictive value.
- ☐ A decrease in sensitivity but no change in specificity

Your answer is incorrect.

By calculating the sensitivity and specificity for each cut-off value, we can produce the following table. Using 4 instead of 3, as one can note from the table, will reduce sensitivity but no change to specificity.

The correct answer is: A decrease in sensitivity but no change in specificity

Question 32

Not answered

Marked out of 1

Using the table below, which reports results of a UK-based study assessing the CAGE to identify people with alcohol dependence, answer questions that follow. What is the optimal cut point in terms of best likelihood ratio for a positive test result?

Select one:

- ☐ 1
- ☐ 4
- ☐ 2
- ☐ 3
- ☐ 0



Your answer is incorrect.

From the attached table, it can be seen that the cut-off of 2 produces the best LR+ value. A rule of thumb : the central (median) value in a set of given cut-off points, wherein sensitivity and specificity appear to have linear but opposite relationships, is the point where the 2 are at their best balance. This, this central point will provide the best LR+ value.

The correct answer is: 2

Question **33**

Not answered

Marked out of 1

Using the table below, which reports results of a UK-based study assessing the CAGE to identify people with alcohol dependence, answer questions that follow. What is the point prevalence of alcohol dependence in the UK?

Select one:

- ☐ 20%
- ☐ Cannot tell from these data
- ☐ 33%
- ☐ 12%
- ☐ 5%



Your answer is incorrect.

To calculate prevalence, we need to know the size of the representative population, and the number of existing cases in this population. Typically, diagnostic assessment studies are based on cross-sectional data collected from a convenient sample of the entire population. Unless it can be assumed with confidence that this sample truly represents the general population, an approximate estimation of true prevalence should not be attempted from studies assessing diagnostic tools.

The correct answer is: Cannot tell from these data

Question **34**

Not answered

Marked out of 1

Using the table below, which reports results of a UK-based study assessing the CAGE to identify people with alcohol dependence, answer questions that follow. What cut-off score would you chose to maximise sensitivity of the CAGE?

Select one:

- ☐ 3
- ☐ 2
- ☐ 1
- ☐ 4
- ☐ 0



Your answer is incorrect.

A cut-off categorises people who score equal to and more than that score in the same section = Test positive group. We can calculate sensitivity and specificity for each of the various cut-off values of the CAGE from the table given in the question. Please keep in mind that the values of the 'total' column in the original table should not be used as such, but must be used in a cumulative fashion according to the cut-off in question. For example, the cut-off of 2 will provide us the following 2x2 table. In this fashion, we can extend the original table with sensitivity, specificity and LR(+) values as below

The correct answer is: 0

Question 35

Not answered

Marked out of 1

A study was conducted to assess the psychometric properties of a 9-item, patient-administered version of the Montgomery-Asberg Depression Rating Scale (MADRS-S). Data from a multicentre, double-blind, 8-week, randomised controlled trial of 278 outpatients diagnosed with Major Depressive Disorder were used to evaluate the validity, reliability and sensitivity to change of the MADRS-S using psychometric methods. From the data collected, a graph was plotted as shown below. Answer the questions below in relation to the graph. [Fantino B, Moore N. The self-reported Montgomery-Asberg depression rating scale is a useful evaluative tool in major depressive disorder. BMC Psychiatry 2009, 9:26]. If the area under the curve for the new test was found to be 0.5, what does it mean?

Select one:

- ☐ The test is half as good as the gold standard test
- ☐ The test can discriminate a depressed from a non depressed person with moderate accuracy
- ☐ The test cannot discriminate a depressed from a non depressed
- ☐ The test has only 50% sensitivity irrespective of the cut-off used
- ☐ The test can discriminate a depressed from a non depressed person with high accuracy

Your answer is incorrect.

As AUC (area under the curve) stands for the accuracy, a test with 50% accuracy will have $AUC=0.5$. In other words this test is only as good as a coin toss.

The correct answer is: The test cannot discriminate a depressed from a non depressed

Question **36**

Not answered

Marked out of 1

A study was conducted to assess the psychometric properties of a 9-item, patient-administered version of the Montgomery-Asberg Depression Rating Scale (MADRS-S). Data from a multicentre, double-blind, 8-week, randomised controlled trial of 278 outpatients diagnosed with Major Depressive Disorder were used to evaluate the validity, reliability and sensitivity to change of the MADRS-S using psychometric methods. From the data collected, a graph was plotted as shown below. Answer the questions below in relation to the graph. [Fantino B, Moore N. The self-reported Montgomery-Asberg depression rating scale is a useful evaluative tool in major depressive disorder. BMC Psychiatry 2009, 9:26]. What is the above graph called?

Select one:

- ☐ Receiver Operator Characteristics curve
- ☐ Galbraith plot
- ☐ Scatter plot
- ☐ Funnel plot
- ☐ Forest plot

Your answer is incorrect.

A receiver-operator characteristics curve is often used to determine cut-off points with best performance in diagnostic tests, and also to compare 2 or more diagnostic instruments.

The correct answer is: Receiver Operator Characteristics curve

Question 37

Not answered

Marked out of 1

A study was conducted to assess the psychometric properties of a 9-item, patient-administered version of the Montgomery-Asberg Depression Rating Scale (MADRS-S). Data from a multicentre, double-blind, 8-week, randomised controlled trial of 278 outpatients diagnosed with Major Depressive Disorder were used to evaluate the validity, reliability and sensitivity to change of the MADRS-S using psychometric methods. From the data collected, a graph was plotted as shown below. Answer the questions below in relation to the graph. [Fantino B, Moore N. The self-reported Montgomery-Asberg depression rating scale is a useful evaluative tool in major depressive disorder. BMC Psychiatry 2009, 9:26]. What does the dotted diagonal line represent?

Select one:

- ☐ It is the curve of a test that discriminates the depressed from non depressed partially
- ☐ It is the median survival curve
- ☐ It is the curve representing the DSM IV diagnosis
- ☐ It is the curve of the test that best discriminates the depressed from non depressed
- ☐ It is the curve of a test that does not discriminate depressed from non depressed

Your answer is incorrect.

The diagonal line represents a test that has only a chance-level ability of discrimination - i.e. one that doesn't differentiate the two groups

The correct answer is: It is the curve of a test that does not discriminate depressed from non depressed

Question **38**

Not answered

Marked out of 1

A study was conducted to assess the psychometric properties of a 9-item, patient-administered version of the Montgomery-Asberg Depression Rating Scale (MADRS-S). Data from a multicentre, double-blind, 8-week, randomised controlled trial of 278 outpatients diagnosed with Major Depressive Disorder were used to evaluate the validity, reliability and sensitivity to change of the MADRS-S using psychometric methods. From the data collected, a graph was plotted as shown below. Answer the questions below in relation to the graph. [Fantino B, Moore N. The self-reported Montgomery-Asberg depression rating scale is a useful evaluative tool in major depressive disorder. BMC Psychiatry 2009, 9:26]. What does 1-specificity represent?

Select one:

- ☐ False positive rate
- ☐ True positive rate
- ☐ Accuracy rate
- ☐ True negative rate
- ☐ False negative rate

Your answer is incorrect.

This refers to the false positivity rate: remember $TN + FP = 100\%$ of cases.

The correct answer is: False positive rate

Question 39

Not answered

Marked out of 1

A study was conducted to assess the psychometric properties of a 9-item, patient-administered version of the Montgomery-Asberg Depression Rating Scale (MADRS-S). Data from a multicentre, double-blind, 8-week, randomised controlled trial of 278 outpatients diagnosed with Major Depressive Disorder were used to evaluate the validity, reliability and sensitivity to change of the MADRS-S using psychometric methods. From the data collected, a graph was plotted as shown below. Answer the questions below in relation to the graph. [Fantino B, Moore N. The self-reported Montgomery-Asberg depression rating scale is a useful evaluative tool in major depressive disorder. BMC Psychiatry 2009, 9:26]. Which cut off point provides the best mixture of sensitivity and some specificity?

Select one:

- ☐ 5
- ☐ 1
- ☐ 2
- ☐ 8
- ☐ 7

Your answer is incorrect.

To determine the point-of-best-performance we must look for the elbow of an ROC curve. At this point, the sensitivity and specificity will be both at a high level reaching a trade-off, though not at their highest values individually.

The correct answer is: 5

Question 40

Not answered

Marked out of 1

You want to carry out a study to determine the diagnostic accuracy of MRI as a screening tool for diagnosing Alzheimer's disease against autopsy confirmed cases. Which of the following study designs is the most suitable?

Select one:

- ☐ Delphi method
- ☐ Case control study
- ☐ Before and after study
- ☐ Cohort study
- ☐ Cross sectional study

Your answer is incorrect.

For evaluation of a screening test, cross-sectional studies are the most suitable. Here, one can start from available autopsy specimens (demented and non-demented), and retrospectively look for MRI reports to construct gold standard vs. screening test (MRI) comparisons. Before and after study is a study in which characteristics of a population or a group of individuals are compared before versus after a particular event or intervention, for example the introduction of a new healthcare service, to gauge what the effects of the event or intervention have been. Delphi is a method of arriving at guidelines from varied expert consensus.

The correct answer is: Cross sectional study

Question 41

Not answered

Marked out of 1

Which one among the following is not generally used to express effect size in meta-analyses?

Select one:

- ☐ Relative risk
- ☐ Odds ratio
- ☐ Likelihood ratio
- ☐ Standardised mean difference
- ☐ Cohen's d

Your answer is incorrect.

The likelihood ratio is a ratio of 2 likelihoods, often used in evaluating diagnostic tests. In meta-analyses absolute measures such as Odds ratio, Relative risk or absolute measures such as standardised mean difference (Cohen's d), Number needed to treat can be used equally to express effect size. If the data is dichotomous (improved vs. non-improvement), then a summary statistic may be a risk ratio. If the data is continuous (like Hamilton depression rating scales) then the summary statistic may be the difference between means.

The correct answer is: Likelihood ratio

Question 42

Not answered

Marked out of 1

Which of the following properties of a screening test increases with increasing disease prevalence in the population?

Select one:

- ☐ Accuracy
- ☐ Specificity
- ☐ Positive predictive value
- ☐ Negative predictive value
- ☐ Sensitivity

Your answer is incorrect.

PPV increases and NPV decreases with increasing prevalence

The correct answer is: Positive predictive value



© 2021 SPMM Course Limited, UK.

Refund Policy | Terms & Conditions



[Return Policy](#) | [Terms & Conditions](#)

